Laksh Gupta

he/him lakshgupta.phy@gmail.com Darmstadt, Germany Indian

Master of Science in Astrophysics student at University of Bonn, Germany, through the scholarship program of the Bonn-Cologne Graduate School of Physics and Astronomy. I studied White Dwarfs using data from the Hubble Space Telescope for my undergraduate thesis supervised by Professor Samyaday Choudhury (Ahmedabad University, India) and in collaboration with Dr. Annalisa Calamida (Space Telescope Science Institute, USA).

Education

Master of Science in Astrophysics, *University of Bonn, Germany* — October 2025 - October 2027

Bachelor of Science (Honors) in Physics, *Ahmedabad University, India* — August 2020 - December 2024

- GPA - 3.32/4
- Minor in Mathematics.
- Relevant Coursework: Quantum Mechanics, Atomic & Nuclear Physics, Condensed Matter Physics, Plasma Physics, Nonlinear Dynamics, Computational Mathematics, Electronics, Quantum Computing, Complex Analysis, Mathematical Statistics, Linear Algebra, Differential Equations, Multivariate Calculus, Advanced Writing

12th Grade, *La Martiniere College, Lucknow, India* — March 2020
Score - 81.2%

10th Grade, *La Martiniere College, Lucknow, India* — March 2018
Score - 84.8%

Research Experience

Research Intern, Astronomy & Astrophysics Group — **June 2025 - August 2025**
Supervisor: Professor Blesson Mathew — *Christ University, Bengaluru, India*

- Working in collaboration with Arun Roy (Indian Institute of Astrophysics) and Shridharan Baskaran (Tata Institute of Fundamental Research), I worked on modelling a Herbig Be star with boundary layer accretion mechanism in RADMC3D to understand forbidden [Ne II] and [Ne III].

Summer Research Fellow — **April 2025 - June 2025**
Supervisor: Professor Blesson Mathew — *Science Academies' Summer Research Fellowship*

- Conducted a detailed theoretical and observational review on accretion mechanisms in Herbig Ae/Be stars, identifying a switch from magnetospheric to boundary layer accretion.
- Prepared a Python pipeline to produce the largest and most complete catalogue of identified Herbig Ae/Be stars from literature.
- Initiated modeling efforts using RADMC3D to study switch in accretion mechanism in Herbig Ae/Be stars, generating SEDs of which match the inclination effects and silicate features in Herbig-like systems.

Research Intern, Astronomy & Astrophysics Group — **May 2024 - September 2025**
Supervisor: Professor Samyaday Choudhury — *Ahmedabad University*

- Working in collaboration with Dr. Annalisa Calamida to refine the star count-crossing time analysis technique developed during the undergraduate thesis.
- This work has merited publication at the **The Astrophysical Journal**.

Research Projects

Undergraduate Thesis: Study of White Dwarfs in the Globular Cluster NGC2808 — **July 2023 - May 2024**
Supervisor: Professor Samyaday Choudhury — *Ahmedabad University*

- Performed star count-crossing time analysis of different evolutionary phases using photometric data of NGC2808 from the HUGS survey.
- Overplotted various BaSTI models (White Dwarf Cooling Models, Isochrones and Evolutionary Tracks) on NUV-optical Color-Magnitude Diagrams to calculate crossing times across different evolutionary phases.
- Created a Python pipeline to systematically analyze stellar populations (White Dwarf Stars, Red Giant Branch Stars, Main Sequence Turnoff Stars) of NGC2808, potentially applicable to other globular clusters.
- *Our results indicate that the White Dwarf Sequence in NGC2808 has no bimodality up to 24 magnitude in F275W.*

Summer Intern — **May 2023 - October 2023**
Supervisor: Professor Navinder Singh — *Physical Research Laboratory, Ahmedabad, India*

- Gained hands-on knowledge of Nuclear Magnetic Resonance (NMR) apparatus.
- Built an NMR apparatus using minimum circuitry and instruments for academic teaching purposes.

Stellar Structure & Evolution — **April 2023 - July 2023**
Supervisor: Professor Samyaday Choudhury — *Ahmedabad University*

- Acquired foundations in stellar structure and evolution - particularly in low-mass stars, radiative transfer and nucleosynthesis in stellar interiors, globular clusters and open clusters, compact objects like white dwarfs, stellar collisions, and binary stars.
- Read *An Introduction to Modern Astrophysics* by Carroll and Ostlie and *Astrophysics for Physicists* by Arnab Rai Choudhuri.

Skills

Astronomy	BaSTI and MIST Models (White Dwarf Cooling Models, Isochrones, Evolutionary Tracks), Optical and Near-IR Hubble Data Analysis, Color-Magnitude Diagram Analysis, Reddening Estimation
Tools	RADMC3D, VizieR, xMatch, TOPCAT, TEX, Tracker
Programming Languages	Python (NumPy, SciPy, Pandas, Astropy, Shapely, Seaborn), MATLAB, Java, R
Communication	Fluent in English (**IELTS** 7 band) and Hindi

Attended Schools & Conferences

43rd Meeting of the Astronomical Society of India — **15th - 19th February 2025**
Poster Presentation — *National Institute of Technology, Rourkela, India*

National School on Space Radiation — **16th - 20th September 2024**
Manipal Center Natural Sciences, MAHE, Manipal, India

School on Solar Physics — **2nd - 6th September 2024**
Physical Research Laboratory, Ahmedabad, India

Other Projects

Profiling a Helium-Neon LASER beam — **December 2022**
Optics Laboratory Project

- Designed the experimental apparatus to profile the He-Ne LASER beam profiling, ensuring precise alignment and calibration of optical components.
- Conducted a detailed error analysis of the experiment.

Need Analysis for Emotional Health and Well-being of students **May 2022 - September 2023**
Supervisor: Professor Shilpa Pandit *Research Project*

- Designed a comprehensive survey to collect data on various aspects of students' mental health ensuring inclusively.
- Cleaned data to remove inconsistencies and converted qualitative responses into numerical values using Python.
- The publication leading to this work is Pandit et al. 2025.

Intrinsic Magnetic Field inside Neodymium Magnets **December 2021**
Electromagnetism Laboratory Project

- Calculated the angular velocity (ω) of the neodymium magnets from the nail-motor experiment apparatus using Tracker software.
- Using ω measurements, the intrinsic magnetic field of the magnets was determined, and a thorough error analysis was conducted.

Newton's Cradle **December 2020**
Classical Mechanics Laboratory Project

- Calculated the translational velocities of the bobs from the Newton's Cradle setup using Tracker software.
- After performing error analysis, translational velocity measurements were utilized to understand conservation of energy and momentum.

Awards & Achievements

- Awarded the **Bonn-Cologne Graduate School of Physics and Astronomy Scholarship** 2025.
- Awarded the **Indian Science Academies' Summer Research Fellowship** 2025.
- **Qualified** IIT-Joint Admission Test for Masters (**IITJAM**) 2025 with an **All India Rank 981** (out of 16,000).

Leadership and Volunteering Activities

- *Peer Tutor* for PHY315 - Atomic & Nuclear Physics (January 2024 - May 2024) and COM102 - Advanced Writing (August 2021 - December 2021)
- Organized a *Math Fest* at Ahmedabad University (November 2023)
- Founded *Ramanujan Math Club* promoting student problem-solving and mathematics at Ahmedabad University (August 2023 - May 2024)
- Represented Physics at Ahmedabad University's *Career Development Centre* (May 2023 - May 2024)
- *Student Mitr* - Mentored 15 incoming students to acquaint them with Ahmedabad University's culture (August 2021 - April 2022)
- *Junior Manager* Outgoing Social Sector - Association Internationale des Étudiants en Sciences Économiques et Commerciales (AIESEC) in Ahmedabad (August 2021 - October 2021)
- Volunteered at Prabhat Foundation during the COVID-19 pandemic (May 2021 - July 2021)